

| Question |     |      |  | Marking details                                                                       | Marks available |     |     |       |       |      |
|----------|-----|------|--|---------------------------------------------------------------------------------------|-----------------|-----|-----|-------|-------|------|
|          |     |      |  |                                                                                       | AO1             | AO2 | AO3 | Total | Maths | Prac |
| 5        | (a) | (i)  |  | 81                                                                                    |                 | 1   |     | 1     | 1     |      |
|          |     | (ii) |  | 20 (2)<br>award (1) for 10 <b>and</b> 30 read from graph if answer is incorrect       |                 | 2   |     | 2     | 2     |      |
|          | (b) | (i)  |  | 122.5 (2)<br>award (1) for $39 + 35.5 + 3(16)$ if answer is incorrect                 |                 | 2   |     | 2     | 2     |      |
|          |     | (ii) |  | $32 / 31.8$ (2) allow ecf from (i)<br>award (1) for $39/122.5$ if answer is incorrect |                 | 2   |     | 2     | 2     |      |
|          | (c) |      |  | $2\text{KCl} + 3\text{O}_2$ both needed                                               |                 | 1   |     | 1     | 1     |      |
|          |     |      |  | Question 5 total                                                                      | 0               | 8   | 0   | 8     | 8     | 0    |